

Flatcar Developer Guide & Migration

Source sections: `devguide/` (renamed from `reference/developer-guides/` in main branch), `coreos-migration/` (renamed from `migrating-from-coreos/`), and `getting-started/learning-series/`.

SDK and Developer Guides (`devguide/`)

The Flatcar SDK is a container-based build environment for building and modifying Flatcar OS images.

Guide	Topic
<code>sdk-bootstrapping</code>	Setting up the build environment
<code>sdk-modifying-flatcar</code>	Patching packages, overlays
<code>sdk-building-production-images</code>	Full image build pipeline
<code>sdk-disk-partitions</code>	Partition layout, GUIDs, USR-A/B scheme
<code>sdk-tips-and-tricks</code>	Common development patterns
<code>kernel-modules</code>	Building and loading out-of-tree kernel modules
<code>release-guide</code>	Cutting a new Flatcar release
<code>integrations</code>	Ansible, Terraform, Packer, and other integrations

Disk partition layout:

Partition	Purpose
EFI System Partition	GRUB bootloader; <code>flatcar/first_boot</code> flag file
USR-A	Active OS image (<code>/usr</code>)
USR-B	Standby OS image (updates written here)
ROOT-A	Active root filesystem
ROOT-B	Standby root filesystem
OEM	OEM vendor tools (as <code>sysex</code> images since 3760.0.0)

Self-Paced Learning Series (`getting-started/learning-series/`)

Four in-depth sessions:

Session	Topic	Prerequisites
Basics and Testing	Local test environment, first Butane config, QEMU VM	None
Advanced Provisioning	Complex deployments, node config management	Session 1
Managing Storage	Additional storage, LUKS encryption at rest	Session 1
Immutability, Updates, Rollbacks	Boot process deep dive, in-place upgrades, automated rollbacks, channels	Sessions 1–2

Migrating from CoreOS Container Linux (`coreos-migration/`)

Flatcar is a drop-in replacement. Key naming changes:

Area	CoreOS	Flatcar
Installer	<code>coreos-installer</code>	<code>flatcar-installer</code>
First-boot flag	<code>coreos.first_boot=1</code>	<code>flatcar.first_boot=1</code>
Config URL	<code>coreos.config.url=URL</code>	<code>ignition.config.url=URL</code>
OEM ID	<code>coreos.oem.id=NAME</code>	<code>flatcar.oem.id=NAME</code>
QEMU fw_cfg	<code>opt/com.coreos/config</code>	<code>opt/org.flatcar-linux/config</code>
VMware guestinfo	<code>coreos.config.data</code>	<code>ignition.config.data</code>
First-boot flag file	<code>/boot/coreos/first_boot</code>	<code>/boot/flatcar/first_boot</code>

Both `coreos.*` and `flatcar.*` kernel parameters are supported in current releases.

In-place update from CoreOS CL:

```
# Run on each CoreOS node
sudo /usr/share/oem/flatcar-update.sh
```

See `coreos-migration/update-from-container-linux.md` for the full procedure including verifying the update server, validating image signatures, and handling rollback if the migration fails.

Roadmap and Future Work (`getting-started/roadmap.md`)

The roadmap file tracks planned Flatcar features and improvements. Key areas as of the `expert-support-refactor` branch:

- Further modularization of the base image via sysext
- Improvements to Nebraska OEM payload delivery
- Additional security hardening toward SLSA Level 4

Flatware Apps (`getting-started/flatware-apps.md`)

Example applications and reference configs for Flatcar Container Linux deployments. Stub file in this branch — content to be added.